Demonstration Kit



Category: **Automotive** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Demonstration Board of Working Model MPFI System with Motorized Control; Transmission / Gearbox Demonstration Kit; Washer and wiper demonstration kit

What it is used for

To demonstrate the operation of automotive systems such as MPFI, transmission or gearbox, and washer-wiper mechanisms using a training demonstration kit. To explain component functions and system operation safely in a controlled learning environment.

Main parts

- Demonstration board or training frame
- Automotive components or working model
- Motor, switches, and control panel, if fitted
- Power supply and wiring
- Labels, diagrams, or indicators
- Protective covers and emergency stop, if fitted

Before you start

1. Identify the specific system and review the kit's operating manual.
2. Inspect wiring, connectors, moving parts, guards, and the power supply.
3. Place the kit on a stable bench with adequate ventilation and lighting.
4. Confirm the rated supply voltage and required protective devices.
5. Keep liquids, loose clothing, and unnecessary objects away from moving parts.

How to use

1. Explain the system layout and identify the main components.
2. Check that all connections and guards are secure before powering on.
3. Connect the specified power source according to the manual.
4. Switch on the kit and demonstrate the controls in the approved sequence.
5. Observe the indicators and explain the system response to each input.
6. For moving mechanisms, keep hands clear and operate only within the permitted range.
7. Switch off and isolate the power before changing connections or inspecting components.
8. Record observations and return the kit to its safe storage condition.

⚠ Safety

- Only trained instructors or supervised learners should operate the kit.
- Never touch exposed live wiring or moving gears while powered.
- Use the emergency stop if unsafe movement or an electrical fault occurs.
- Do not bypass guards, interlocks, or protective devices.
- Follow the manual for fuel-system, motor, and electrical demonstrations.

Care & storage

- Switch off and disconnect power after use.
- Clean the board and components with suitable materials.
- Inspect cables, connectors, guards, and controls regularly.
- Report faults and store the kit in a dry, secure area.

Common mistakes to avoid

- Powering on before checking wiring and guards.
- Using an incorrect supply voltage or connection.
- Touching moving or live components during operation.
- Changing wiring while the kit is energized.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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